



Qualcomm Technologies, Inc.

Qualcomm Linux Storage Guide

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Contents

- 1 Overview 3
- 2 Features 4
- 3 Configure 5
 - 3.1 Kernel configuration 5
 - 3.2 Power management states 9
 - 3.3 Bootloader/UEFI device tree 10
 - 3.4 Linux device tree 11
- 4 Customize 13
- 5 Debug 15
 - 5.1 Debug UFS issues 15
 - 5.2 Debugging with sysfs or debugfs nodes 17
- 6 References 19

1 Overview

Understanding the storage features and device types supported on the Qualcomm® QCS6490 platform enables developers to configure, customize, and deploy storage capabilities in an optimal, efficient way.

- Universal flash storage (UFS) versions 2.1/2.2 and 3.1.

NOTE Universal flash storage (UFS) is an open standard set by the Joint Electron Device Engineering Council (JEDEC). It is a high-performance storage interface designed to minimize power consumption. Its high-speed serial interface and optimized protocol enable significant improvements in throughput and system performance.

- SD card version SD v3.0, 4-bit

Storage is used for the following use cases on this platform:

- **Boot from primary storage device**

Store the required images to boot up the Qualcomm QCS6490 platform by reading the images (XBL, AOP, TrustZone, kernel, and so on) from UFS

- **UFS has a partition that saves user-specific data**

For example, videos, photos, documents, and so on.

- **Support read only logical unit number (LUN)/region**

Used for storing images such as TZ, CDT, and so on.

NOTE LUN is an independent processing unit on a UFS device.

- **Stores data in an authenticated, access-controlled, replay-protected manner in RPMB LUN/partition**

Replay protected memory block (RPMB) is a well-known LUN (WLUN) in UFS.

2 Features

UFS features

The following table lists the **UFS** features supported on this platform.

Feature	Description
Number of lanes	Only 2 lanes is supported
Gear	High-speed (HS) Gear 1, 2, 3, and 4 are supported
Low-power mode	Supports Link Hibernate and device Sleep modes . See UFS power management states for more information.
Embedded UFS	Only the embedded UFS card is supported. Hot plug is not supported.

SD card features

The following table describes the **SD card** features supported on this platform.

Feature	Description
Number of bits	SD card can operate in 4-bit mode
Speed	SDR50, SDR104, and DDR50
Hot-plug capability	Hot plug is supported for SD cards.

3 Configure

This section explains various storage configurations that can be used by developers. These configurations include kernel configurations for UFS functionality, and device tree parameters.

3.1 Kernel configuration

A default UFS kernel configuration enables UFS functionality.

The following table provides a list of kernel configurations for UFS that can be modified by the developer. The default values of these configurations included in the build are described below.

Table 3-1 UFS kernel configuration

Configuration	Enabled (Yes/No)	Comments
CONFIG_UFS_FAULT_INJECTION	No	Injects the failure command for debugging
CONFIG_SCSI_UFSHCD	Yes	Enables the UFS host control driver in the kernel
CONFIG_SCSI_UFSHCD_PLATFORM	Yes	States that the UFS controller is supported by the platform bus
CONFIG_SCSI_UFS_QCOM	Yes	Enables a QTI-specific addition to access the hardware, that is, PHY configuration and vendor-spec
CONFIG_PHY_QCOM_QMP	Yes	Supports UFS QMP PHY driver
CONFIG_SCSI_UFS_BSG	Yes	Enables UFS BSG device node
CONFIG_SCSI_UFS_CRYPTO	No	UFS crypto engine support

A default SDHC kernel configuration enables SD card functionality.

The following table provides a list of kernel configurations for SD card that can be modified by the developer. The default values of these configurations included in the build are described below.

Table 3-2 SDHC kernel configuration

Configuration	Enabled (Yes/No)	Comments
CONFIG_MMC	Yes	Enables MMC driver
CONFIG_MMC_DEBUG	No	Enables driver debug messages
CONFIG_MMC_SDHCI	Yes	Detects SDHCI driver
CONFIG_MMC_CQHCI	Yes	Enables command queue host control interface
CONFIG_MMC_SDHCI_MSM	Yes	Enables MSM SDHCI driver
CONFIG_MMC_BLOCK	Yes	Enables MMC block driver

The kernel configurations as per the preceding table are enabled by default in `<workspace_root_path>/sources/kernel/kernel_platform/kernel/arch/arm64/configs/defconfig` and `qcom_defconfig` files.

3.2 Power management states

UFS and the SD card offer multiple power states to optimize power consumption.

UFS power management states

Developers can alter these states as per requirement. However, it is to be noted that altering of the states may also lead to power consumption variation..

State	Description
Auto-Hibernate	Auto-hibern8 (AH8) automatically transitions the phy link into Hibernate state after 150 ms of idle duration. To check AH8 idle duration value (in micro sec), run the following ssh command on the host machine: <pre>cat /sys/devices/platform/soc@0/1d84000.ufs/auto_hibern8</pre> Note: This value can be decreased to reduce power consumption.
Clock-gating	When UFS clocks are scaled up to maximum frequency due to the use case, clocks are not gated. If clocks are scaled down to optimal frequency, clocks are gated after 10 ms of idle duration. To check clock gating status, run the following ssh command on the host machine: <pre>cat /sys/devices/platform/soc@0/1d84000.ufs/clkgate_enable</pre> Clock gating is enabled by default. To debug UFS issues related to clock gate/ungate, clock gating can be disabled using the following ssh command on the host machine: <pre>echo 0 > /sys/devices/platform/soc@0/1d84000.ufs/clkgate_enable</pre>
Clock scaling	Once the UFS load exceeds the <code>upthreshold</code> value, the clock frequency is set to the maximum. Clock scale down happens once the load is below the threshold. UFS load is determined by the <code>upthreshold</code> status, run the following ssh command on the host machine: <pre>cat /sys/devices/platform/soc@0/1d84000.ufs/clkscale_enable</pre> Clock scaling is enabled by default. Disable clock scaling can result in increased power consumption. To debug any issue related to clock scaling, run the following ssh command on the host machine: <pre>echo 0 > /sys/devices/platform/soc@0/1d84000.ufs/clkscale_enable</pre>
Runtime suspend/resume	UFS enters the runtime suspend state after 3 seconds of inactivity. Resume can happen after 3 seconds. Not configurable.
System suspend/resume	UFS enters system suspend with the suspend event. UFS resume happens with the system resume event. Not configurable.

SD card power management states

State	Description
Runtime suspend/resume	The SD card enters the runtime suspend state after 50 milliseconds of inactivity. Resume happens after 50 milliseconds. Enabled always and not configurable.
System suspend/resume	The SD card enters system suspend with the suspend event. SD card resume happens with the system resume event. Not configurable.

3.3 Bootloader/UEFI device tree

The device tree specifies the UFS host configuration parameters such as number of gears, number of lanes, rate value, timeout values, etc.

UFS

The device tree is located at <workspace_root_path>\boot_images\boot\Settings\Soc\Kodiak\Core\Storage\UFS\ufs.dtsi.

The UFS device tree node in UEFI is described below.

Device tree node/key	Subnode	Device tree value	Notes
init_speed_params	EnableHighSpeed	1	Enables HS mode
	NumGears	4	Max gear to be used.
	NumLanes	2	Number of lanes used in UFS
	Rate	2	▪ 1 = Rate A ▪ 2 = Rate B
perf_speed_params	EnableHighSpeed	1	To enable HS mode while in perf mode
	NumGears	4	Max gear to be used in perf mode.
	NumLanes	2	Number of lanes used in UFS in perf mode
	Rate	2	Series in perf mode ▪ 1 = Rate A ▪ 2 = Rate B
timeout_values	fDeviceInitTimeoutUs	2500000	Timeout (μs) during device init
	UTRDPollTimeoutUs	30000000	Timeout (μs) for UTRD
BatteryThresholdMv		3600	Battery threshold (in milli volts) to move to Gear1
LinkStartupRetryCount		5	Retry count for Link startup
FUA_Value		1	Forced unit access
EnableLogging		0	Enables DT serial debug logging in loader ▪ 1 - Enable ▪ 0 - Disable
RefClock		19200000	UFS controller reference clock 19.2MHz
MphyInitTable			MphyInit table
MphyInitEndTable			MphyInitEnd table
MphyLaneInitTable			Mphy 2 Lane Init table

Any change in the `.dtsi` should be compiled into `xbl_config.elf` and the image reflashed on to the device.

NOTE `MphyInitTable`, `MphyInitEndTable`, and `MphyLaneInitTable` are used for UFS phy initialization and these values are tuned by Qualcomm. For debugging/tuning any phy-level issue, these values should be changed with guidance from the Qualcomm team.

NOTE The default values are for UFS 3.1 part. For UFS 2.x part, `NumGears` value should be 3. Other parameters need not be changed for UFS 2.x part.

3.4 Linux device tree

The UFS device tree nodes define properties used for storage.

UFS

The SoC DTSI node specifies register address space, clocks, interrupts, and reset information. The platform DTSI node specifies power supply, voltage, and current levels. The `Ufs_mem_hc` node describes the on-chip UFS host controller and this node is in the `<workspace_root_path>/sources/kernel/kernel_platform/kernel/arch/arm64/boot/dts/qcom/sc7280.dtsi` file.

NOTE These parameters are only for information purposes and should not be changed by the developer.

Property	Description
Compatible	For Qualcomm SoCs, must contain strings such as "qcom, ufshc".
Interrupts	Interrupt mapping for UFS host controller IRQ
Reg	UFS host registers address mapping
Phys	<code>phandle</code> to UFS PHY node
lanes-per-direction	Specifies the number of lanes available per direction. Either 1 or 2.
clock-names	List of clock input name strings
Clocks	List of <code>phandle</code> and clock specifier pairs
freq-table-hz	Array of <code><min max></code> operating frequencies stored in the same order as the clocks proper
reset-gpios	A <code>phandle</code> and GPIO specifier denoting the GPIO that is connected
Resets	Reset node register

The `Ufs_mem_phy` node describes on-chip UFS PHY hardware and this node is in `<workspace_root_path>/sources/kernel/kernel_platform/kernel/arch/arm64/boot/dts/qcom/sc7280.dtsi` file.

Property	Description
Compatible	Compatible string such as "qcom,qmp-ufs-phy" needs to be specified.
#phy-cells	Property should be set to 0
Reg	Should contain PHY register address space
reg-names	Indicates various resources passed to driver (via reg property) by name. The required reg-names is <code>phy_mem</code> .

Property	Description
lanes-per-direction	Number of lanes available per direction; either 1 or 2.
clock-names	List of clock input name strings
Clocks	List of phandle and clock specifier pairs
vdda-phy-supply	phandle to main PHY supply for analog domain
vdda-pll-supply	phandle to PHY PLL and Power-Gen block power supply
Resets	Specifies the PHY reset in the UFS controller

For more information on DTS parameters, refer to: [/kernel_platform/msm-kernel/Documentation/devicetree/bindings/ufs/qcom,ufs.yaml](#).

4 Customize

This section explains how to customize the storage features including adding a new partition and mounting the SD card to a specific path.

File system

ext4 is an open-source file system that is used on data partitions on the Qualcomm Linux platform.

ext4 is a [journaling file system](#) for Linux. The ext4 file system can support volumes with sizes in theory up to 64ZB and single files with sizes up to 16TB with the standard 4KB block size.

An extent is a range of contiguous physical blocks, improving large-file performance and reducing fragmentation. A single extent in ext4 can map up to 128 MiB of contiguous space with a 4 KB block size.

ext4 does not limit the number of subdirectories in a single directory, except by the inherent size limit of the directory itself.

For more information, see [An introduction to Linux's EXT4 filesystem | Opensource.com](#)

Data partition is mounted at `/mnt/overlay`.

For example:

```
/dev/sda9 on /mnt/overlay type ext4
(rw,relatime,rootcontext=system_u:object_r:etc_t:s0,seclabel,stripe=128,inline
crypt)
```

To mount `/dev/sda9` to `/data` use the following commands:

```
/ # mkdir /home/root/data
/ # mount -o rw,remount /
/ # mount -t ext4 /dev/sda9 /home/root/data
```

NOTE If you encounter any issues related to SELINUX, please use the following command.

```
mount -o rootcontext=system_u:object_r:etc_t:s0 -t ext4 /dev/sda9 /
home/root/data
```

Execute the `mount` command to retrieve the exact `rootcontext=<>` value related to the `/mnt/overlay`.

Mount the SD card to a specific path

If the SD card is not mounting automatically, please use the following commands to manually remount it to a specific path.

1. Remount the root partition as read-write:

```
sh-5.1# mount -o rw,remount /
```

2. Create an SD card directory:

```
sh-5.1# mkdir sdcard
```

3. Mount the SD card to the previously created path (*/sdcard*):

```
sh-5.1# mount -t vfat /dev/mmcblk1p1 /sdcard
```

4. To verify whether the SD card is mounted, check the available space using:

```
sh-5.1# df -h
```

```
..
```

```
/dev/mmcblk1p1 7.4G 101M 7.4G 2% /sdcard
```

NOTE If the SD card has multiple partitions, follow step 3 to mount each partition.

Preferred Vendor List (PVL) memory part information

All UFS parts supported on the Qualcomm SoC are qualified in the PVL validation. Select the applicable part from the [Memory list](#).

5 Debug

This section describes different errors from the storage driver and provides guidance on debug to resolve such errors.

5.1 Debug UFS issues

This section describes how to debug different UFS issues.

UFS device issue

When a UFS device issue occurs *cmd pending in the device* or *no response from device* error log recorded in the kernel log as shown below. Check with the UFS vendor for more information if this issue occurs.

Example 1:

```
[ 82.928056][01-01 00:02:05] ufshcd-qcom 1d84000.ufshc: ufshcd_abort: Device
abort task at tag 1
[ 82.928059][01-01 00:02:05] sd 0:0:0:4: [sde] tag#1 CDB: Read(10) 28 00 00
00 44 49 00 00 20 00
[ 82.936477][01-01 00:02:05] ufshcd-qcom 1d84000.ufshc: UPIU[1] - issue time
50955434 us
[ 82.936479][01-01 00:02:05] ufshcd-qcom 1d84000.ufshc: UPIU[1] - complete
time 0 us
[ 82.936567][01-01 00:02:05] ufshcd-qcom 1d84000.ufshc: ufshcd_abort: cmd
pending in the device. tag = 1
```

Example 2:

```
[ 1036.008147] [1230_22:00:49]@0 ufshcd-qcom 1d84000.ufshc: ufshcd_abort:
Device abort task at tag 2 [ 1036.008510] [1230_22:00:49]@0 sd 0:0:0:0: tag#2
CDB: opcode=0xfa (vendor) fa 02 01 48 07 79 ae 03 50 00
[ 1036.120287] [1230_22:00:50]@1 ufshcd-qcom 1d84000.ufshc: ufshcd_abort: no
response from device. tag = 2, err -110.
```

PA/DL error

In the kernel log, when we see *pa_err_cnt_total* and *dl_err_cnt_total* as non-zero then we can consider this as a PA/DL error issue. This issue indicates that the signal quality between the SoC and the UFS device is not strong. Typically, resolving such issues involves collaboration among hardware, software, and the UFS vendor.

Use the debugfs path `/sys/kernel/debug/ufshcd/1d84000.ufs/stats` to see the PA/DL error count.

debugfs should be mounted using the following command. This command will make the debugfs filesystem accessible for debug nodes.

```
mount -t debugfs none /sys/kernel/debug
```

Example:

```
[31709.438001] ufshcd-qcom 1d84000.ufshc: pa_err_cnt_total=5086,
pa_lane_0_err_cnt=3250,
pa_lane_1_err_cnt=1405, pa_line_reset_err_cnt=431
[31709.438008] ufshcd-qcom 1d84000.ufshc: dl_err_cnt_total=4488,
dl_nac_received_err_cnt=289, dl_tcx_replay_timer_expired_err_cnt=258
[31709.438015] ufshcd-qcom 1d84000.ufshc:
dl_afcx_request_timer_expired_err_cnt=0,
dl_fcx_protection_timer_expired_err_cnt=0, dl_crc_err_cnt=18
```

UFS power consumptions issue

This section explains the debug steps to troubleshoot excess power consumption issues due to active/sleep power states.

Active state

During active state, if UFS power consumption is high, use the following steps to debug.

1. Check how often the issue occurs.

Use the following ssh command to check current clock frequency:

```
# cat /sys/devices/platform/soc@0/1d84000.ufs/devfreq/1d84000.ufs/cur_freq
```

If the frequency is maximum, skip to Step 3.

2. Check whether any UFS-related feature (such as WriteBooster) has been enabled.
3. Check whether clock scale is disabled using the following ssh command on the host machine:

```
#cat /sys/devices/platform/soc@0/1d84000.ufs/clkscale_enable
```
4. Check the UFS clocks frequency and UFS rails at the point of issue.
5. The issue may occur after device bootup due to initial configuration or during suspend/resume due to an unexpected devfreq event.

Sleep state

During sleep state, if UFS power consumption is high, use the following steps to debug UFS suspend-related issues.

1. Check the Power management state using sysfs nodes `link_state`, `curr_freq`, and `runtime_status` and PM level (PM_LVL 3 has higher power consumption than 5).
2. Check the UFS state using sysfs node `dev_pm` to determine whether UFS is in sleep mode.
3. Check whether the HPB and write booster features are enabled because it causes high power consumption.
4. Check the UFS VCC/VCCQ/VCCQ2 timing from the hardware side if an issue occurs.

Disable UFS SMMU

System Memory Management Unit (SMMU) is a memory management unit (MMU) that connects a direct-memory-access-capable (DMA-capable) I/O bus to the physical memory.

Each device is assigned a unique stream ID (SID). UFS controller SID is 0x80 on this platform.

If there is an unhandled SMMU fault from UFS SMMU as shown in the log below, then UFS SMMU can be disabled.

```
arm-smmu 15000000.apps-smmu: Unhandled arm-smmu context fault from
1d84000.ufshc!
```

UFS SMMU can be disabled with the following change in `ufshc_mem` node in `<workspace_root_path>/sources/kernel/kernel_platform/kernel/arch/arm64/boot/dts/qcom/sc7280.dtsi` file.

```
- qcom,iommu-dma = "fastmap";
+ qcom,iommu-dma = "bypass";
```

5.2 Debugging with sysfs or debugfs nodes

There are various sysfs and debugfs nodes that can be used to identify the status or update the state.

debugfs should be mounted using the following command. This command makes the debugfs file system accessible for debug nodes.

```
mount -t debugfs none /sys/kernel/debug
```

To obtain UFS state information

Required UFS state information	Command
Check PA/DL error count	<code>cat /sys/kernel/debug/ufshcd/1d84000.uf</code>
Read UFS gating delay, in ms	<code>cat /sys/devices/platform/soc@0/1d84000</code>
Read UFS clocks current frequency at given point	<code>cat /sys/devices/platform/soc@0/1d84000</code>
Read UFS I/O scheduler details	<code>cat /sys/devices/platform/soc@0/1d84000</code>
Read UFS min frequency during clock scale down	<code>cat /sys/devices/platform/soc@0/1d84000</code>
Read UFS max frequency during clock scale up	<code>cat /sys/devices/platform/soc@0/1d84000</code>
Read UFS device state ACTIVE or SLEEP	<code>cat /sys/devices/platform/soc@0/1d84000</code>
Read UFS gear information such as HS_G3, HS_G4	<code>cat /sys/devices/platform/soc@0/1d84000</code>
Read UFS number of lanes	<code>cat /sys/devices/platform/soc@0/1d84000</code>
Read UFS link state, ACTIVE or HIBERN8	<code>cat /sys/devices/platform/soc@0/1d84000</code>

Required UFS state information	Command
Read UFS rate information such as Rate A or B	<code>cat /sys/devices/platform/soc@0/1d84000</code>
Read UFS device descriptor information	<code>ls /sys/devices/platform/soc@0/1d84000.</code>
Read UFS health descriptor information	<code>ls /sys/devices/platform/soc@0/1d84000.</code>
Read UFS runtime status information active or suspended	<code>cat /sys/devices/platform/soc@0/1d84000</code>
Find and/or modify the SPM/RPM level	<code>cat /sys/devices/platform/soc@0/1d84000</code> <code>cat /sys/devices/platform/soc@0/1d84000</code>
To obtain SD card state information	
Required SD card state information	Command
Read SD card error stats information	<code>cat /sys/kernel/debug/mmc1/err_stats</code>
Read SD card latest error state details	<code>cat /sys/kernel/debug/mmc1/err_state</code>
Check the current running clock details	<code>cat /sys/kernel/debug/mmc1/clock</code>
Read the SD card runtime status information (active or suspended)	<code>cat /sys/devices/platform/soc@0/8804000</code>
Read SD card auto suspend delay time	<code>cat /sys/devices/platform/soc@0/8804000</code>
Read SD card runtime active time	<code>cat /sys/devices/platform/soc@0/8804000</code>
Read SD card runtime suspend time	<code>cat /sys/devices/platform/soc@0/8804000</code>
Read SD card uevent details	<code>cat /sys/devices/platform/soc@0/8804000</code>

6 References

Table 6-1 References

Reference	Definition
PCAT	Qualcomm Product Configuration Assistant Tool

Table 6-2 Acronyms

Acronym or term	Definition
DDR	Double data rate
eMMC	Embedded multimedia card
eUFS	Embedded UFS
LUN	Logical unit number
OTP	One-time provisioning
SDC	Secure digital card
SDCC	Secure digital card controller
UFS	Universal flash storage

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